

Double-Verb Constructions, Markedness, and a Fake Co-ordination*

Guy Carden and David Pesetsky
Yale University

1. Introduction

- (1) Come live with me and be my love... Marlowe

In this paper we discuss double-verb constructions like (1) and the closely related "Come and live with me." The first verb of the V_1V_2 construction is limited to the motion verbs come, go, and, for some speakers, run (2); the V_1 and V_2 allows a wider range of V_1 , and the list in (3) is probably not complete.

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|------------|-----------------|-------------------|
| (2) VV^1 | (3) V and V | |
| come live | come and live | try and catch |
| go visit | go and visit | be sure and shave |
| run get | run and get | hurry up and get |

In section 2, we argue that these double-verb constructions appear only when both verbs are "bare verb stems," and, in particular, that the criterion involves a lack of morphological marking rather than merely phonological identity with the bare stem. This conclusion is of some interest for marking theory, since it implies a contrast between "morphologically unmarked" and "marked with a phonologically zero marking." In addition, our analysis implies that the English "unmarked present tense" is morphologically as well as phonologically unmarked, with consequences for the analysis of the auxiliary.

In sections 3-6 we discuss the underlying structure of these constructions. We do not reach any firm conclusion about what that underlying structure is, but we give a number of arguments about what it is not. In particular, we show that the and of the V and V structure is a fake and rather than a real co-ordinate construction. The evidence suggests that both double-verb constructions come from a common complement-like structure, but we argue that a simple derivation from an infinitive complement will not work.

2. The bare-stem condition

In this section we concentrate on the VV construction, but we claim that the same results hold for V and V . This will become clear in sections 3 and 4, where we discuss the semantics and the problem of controlling for irrelevant readings.

For the VV construction, you get a surface paradigm like (4):

- (4) a. Infinitive: John managed to go visit Harry last week.
 b. Imperative: Go visit Harry tomorrow!
 c. Modals: John will go visit Harry tomorrow.
 d. *Past: *John went visit Harry yesterday.
 *John go visited Harry yesterday.
 *John went visited Harry yesterday.²
 e. *Perfect: *John has gone visit Harry already.
 *John has gone visited Harry already.²
 f. *ing *John is going see Harry tomorrow.
 *Going see Harry was probably a mistake.

The apparent generalization is that you get the VV construction only if both Vs are bare stems, with no affixes. This is confirmed when we look at questions and negatives:

- (5) a. Did John go visit Harry yesterday? Do+PAST NP Stem
 b. *Has John gone visit Harry already? have+S NP Stem+PP
 (6) a. John didn't go visit Harry yesterday. do+PAST NEG Stem
 b. *John hasn't gone visit Harry yet. have+S NEG Stem+PP

If the PAST affix is separated from the stem by Subject-Aux Inversion (5a) or NEG-Placement (6a), the bad past tense examples (*4d) become good. No such improvement appears in the perfect (*4e, *5b, *6b), since the PP affix remains on the stem. This three-way contrast -- affirmative past bad (*4d), past question OK (5a), perfect question bad (*5b) -- confirms our generalization: The violation must involve the presence of an affix late in the derivation, and not some deeper interaction of tense, question/negation, and the VV structure.

There are two ways to state this generalization: We can say that each of the Vs must be phonologically identical to its bare stem, or we can say that it must be the bare stem, with no morphological affixes. Conveniently, we have a test case: The past participle of come is phonologically identical to the bare stem. When we construct the examples, we see that the paradigm of come V is identical to that of go V:³

- (7) a. *Past: *John came live with us.
 b. *Perfect: *John has come live with us.
 (8) a. Did John come live with you?
 b. *Has John come live with you?
 (9) a. John didn't come live with us.
 b. *John hasn't come live with us.

The crucial contrast is between the good past-tense questions and negatives (8a, 9a), where come is morphologically a bare stem, and the bad perfect-tense questions and negatives (8b, 9b), where the surface come is morphologically stem+PP. We conclude that the generalization must be stated in terms of morphological marking, and that phonological shape is irrelevant.

If we accept this conclusion, it follows that the stem-like forms in infinitives, imperatives, and modal constructions are in fact bare stems with no morphological marking. This conclusion may not be controversial, but let's look at what happens in the present

tense. The present is usually described as unmarked semantically, but it shows a morphological tense marker in standard analyses. The present tense paradigm shows a sharp contrast between the 3rd person singular, which is bad (*10), and all other persons and numbers, which are OK (11):

- (10) a. *John goes visit Harry every afternoon.
 goes visits
 go visits
- b. *If John comes visit us, we'll take him sailing.
- (11) a. I go visit Harry every Thursday.
 b. If you come visit us, we'll take you sailing.
 c. If we go ask Harry, will he be able to tell us?
 d. You two always go visit Harry on Fridays, don't you?
 e. They come visit us here once a year.

As with our previous examples, we can use questions and negatives to confirm that the problem with (*10) is a violation of the bare-stem condition:

- (12) a. Does John go visit Harry on weekends?
 b. Does John come visit you in the summer?
 c. John doesn't go visit Harry every day, does he?
 d. John doesn't come visit us very often.

Further confirmation comes from examples with be sure and V: Since to be has marked agreements in all persons of the present tense, the bare-stem condition predicts that the V and V construction will be entirely blocked in the present. That is correct:

- (13) a. Be sure and visit Harry tomorrow.
 b. Tom should be sure and visit Harry tomorrow.
 c. *I am sure and visit Harry every Tuesday.
 d. *They are sure and visit Harry once a week.⁴

These examples confirm that there is no interaction between tense person as such and the double-verb construction; all that matters is whether a given tense/person combination is morphologically marked.

This conclusion has implications for the analysis of the present tense: If our bare-stem generalization is correct, the semantically unmarked present tense must also be morphologically unmarked except in the 3rd person singular (and all forms of be). If there were a phonologically zero tense marker, as in standard analyses, we would expect (11) to act like the examples with the perfect participle come.⁵ In fact, however, the examples with come plus phonologically zero PP are all bad (*7b, *8b, *8b), and the examples with present tense and no visible marker are all good (11).⁶ This contrast shows that the present-tense examples must have no marking, rather than a phonologically zero marking: the present tense is morphologically as well as semantically unmarked.⁷

This conclusion causes technical problems for the classical Syntactic Structures auxiliary analysis: If there is no present-tense morpheme except in the 3rd person singular, then there will

be no lonely tense to trigger do-Support in examples like The boys don't like apples. Various ad hoc patches are possible, but it is worth pointing out that the problem does not arise in an analysis with do-Deletion rather than do-Support.

3. Underlying structure: Evidence for a fake and

In the previous section we used the surface bare-stem condition to build an argument about the unmarked present tense; in this section we begin to attack the problem of the underlying structure of double-verb constructions. We start with the easy case try and V, and present arguments that this and is a "fake" that does not correspond to an underlying co-ordinate conjunction.

A surface string try and V is typically ambiguous between what we will call "fake-and" and "real-and" readings:

- (14) a. John will try and catch Harry. (ambiguous)
 b. John will try to catch Harry.
 c. John will try (to do X), and he will catch Harry.

The fake-and reading is a close paraphrase of the complement structure (14b); note that it does not entail that John will succeed in catching Harry. The real-and reading is a reduced form of an underlying structure like (14c); it requires a deleted complement after try, and it does imply that John will succeed in catching Harry.

3.1 Phonology

- (15) a. John will try [n] catch Harry. (fake-and)
 b. John will try, and catch Harry. (real-and)

The fake-and reading normally takes a much-reduced [n] for and; the real-and reading normally has a pause at the site of the deleted complement followed by a full and. This phonological difference is usually enough to disambiguate in speech.

3.2 Syntax

There are three syntactic contrasts between real-and and fake-and readings:

A. The same bare-stem condition that applies to the VV construction also applies to try and V:

- (16) a. ?John tried and caught me.
 b. John didn't try and catch me.
 c. Did John try and catch you.
 a'. *John tried and catch me.

(16a), which violates the bare-stem condition, has only a real-and reading, and seems un-natural because of the lack of an antecedent for the deleted complement of try. If we put a tense morpheme on only one V (16a'), the result has neither reading and is completely ungrammatical. If we separate the tense affix from the stem by NEG-Placement (16b) or Subject-Aux Inversion (16c), we get a good fake-and reading.

B. As Faraci (1970) points out, fake-and structures are not subject to the Co-ordinate Structure Constraint:

- (17) a. Q: Was John well enough to run the 1500?
 A: He tried, and caught Bill in the last 200m.
 b. *Bill is the man that John tried, and caught in the last 200m.
 c. Bill is the man that John will try and catch.

Relative Clause Fronting can remove an object of V_2 in a fake-and structure (17c); but if we use intonation and the bare-stem condition to force a real-and reading (17a), then Relative produces a normal CSC violation (*17b).

This surface exception to the CSC is not necessarily linked to the double-verb construction we are studying, since other surface ands that are not semantically co-ordinate also are not subject to the CSC:

- (18) a. ?OK This is the girl that John should move to Boston
 and live with.
 b. "What's the name of the stuff that you hang up at
 Christmas and people walk under it and kiss you?"
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C. Parentheticals appear in the middle of real-and co-ordinations, but not in the fake-and structure:

- (19) a. John will, unfortunately, try and catch me.
 b. John will try, unfortunately, and catch me.
 c. John will try and, unfortunately, catch me.
 d. Tom, Dick, and, unfortunately, Harry assembled at 12.

(19a) has a fake-and reading, but (19bc) have only the real-and reading that implies that John will succeed in catching me.

3.3 Semantics

The semantic contrasts between the fake-and and real-and readings can be seen most sharply in a discourse context. Consider first the discourse effect of the deleted complement in the real-and reading:⁸

- (20) Q: What happens in the next act?
 a. A: John will try and kiss Mary. (coherent)
 b. A: #John tries and kisses Mary. (needs more context)
 c. A: #John will try, and he will kiss Mary.

(20a) gives a coherent discourse on a fake-and reading; if we force a real-and reading by violating the bare-stem condition (20b), the discourse becomes incoherent like the unreduced conjunction (20c). In both the real-and structures, we need more context to supply the deleted complement -- What does John try to do? If that context is lacking, we can force a backwards anaphora reading and deduce that John tries to kiss Mary; but, as the familiar conditions on backwards anaphora predict,⁹ the resulting discourse is not fully acceptable.

We get a parallel disambiguation in discourses where a continuation specifies the success or non-success of V_2 :

- (21) Q: Should Snoopy and his squadron intercept the Red Baron?
 a. A: If they try, and hit the Red Baron, his plane will crash.
 b. A: If they try and hit the Red Baron, he'll shoot them down for sure.

In (21a) we have the real-and reading, with a continuation implying that Snoopy and company succeed in hitting the Red Baron; in (21b) we have a continuation implying non-success, and the fake-and reading is forced. If we force the real-and reading by violating the bare-stem condition, a continuation that implies non-success is incoherent (#22b):

- (22) a. A: Every time they've tried, and hit the Red Baron, his plane has crashed.
 b. A: #Every time they've tried and hit the Red Baron, they missed and he shot them down.

A third contrast involves a selectional restriction on V_2 :

- (23) Do you think John can cross the river on that narrow plank?
 a. A: John will try, and fall in the river.
 b. A: John will try [n] fall in the river.
 c. A: John will try to fall in the river.

With the real-and reading, V_2 is unconstrained; in particular, a non-self-controllable like fall is perfectly natural. If we enforce the fake-and reading with intonation (23b), the result is semantically marked: we must interpret fall as self-controllable. This implies a selectional restriction on V_2 of try and V that is identical to the selectional restriction on infinitive complements of try; note that (23c) has the same semantically marked interpretation as (23b).

4. Fake and in the other V and V structures

In this section we attempt to extend the arguments of section 3 to show that there is also a fake and in come/go/run/hurry up/be sure and V. The extension to be sure and V is straightforward, but the other V_1 have different semantic properties that make it impossible to reconstruct two of the arguments of section 3.3. In addition, these verbs permit irrelevant readings that have to be distinguished to apply the syntactic tests.¹⁰ Some of the judgements required are quite subtle, but we claim nonetheless that the evidence supports an analysis in which all of these V and V share the fake-and and the bare-stem constraint that we demonstrated for try and V.

4.1 be sure and V

All the arguments of section 3 apply directly to be sure and V with two exceptions: the argument based on the success/non-success contrast (see 4.2) and the bare-stem argument of section 3.2:

- (24) a. John will be sure and visit Harry tomorrow.
 b. *John was sure and visit(ed) Harry yesterday.
 c. *Was John sure and visit(ed) Harry yesterday?
 d. *John wasn't sure and visit(ed) Harry yesterday.¹¹

The fake-and reading is blocked in (24b) as expected (and the real-and reading requires enough context that the example appears *d in isolation); but the fake-and reading fails to reappear in questions and negatives (*24cd). The badness of (*24cd), however, is predictable from the interaction of the bare-stem condition with NEG-Placement and Subject-Aux Inversion, which will treat be as and auxiliary. The non-parallelism between be sure and try in these examples actually provides additional indirect confirmation for the bare-stem condition.

4.2 Implicative V₁

The fake-and reading of try and V is non-factive;¹² (24a) does not imply (24b). We used this to build the argument of (21,22). This is not true for any of the other V₁: both the fake-and and real-and readings of (25a) imply (25b), and it follows that we cannot reconstruct the success/non-success argument of (21,22) for any of these V₁.

- (24) a. John will try and catch Harry.
 b. John will catch Harry.
 (25) a. John will come and visit us tomorrow.
 b. John will visit us tomorrow.

This eliminates one of our arguments, but does not cast doubt on the argument as a whole; This contrast between try and the others is predictable from the fact that try is non-factive in general, while the others are implicative:

- (24') a. John will try to catch Harry.
 (25') a. John will come to visit us tomorrow.
 c. John won't come to visit us tomorrow.
 d. John won't come and visit us tomorrow.

The infinitive complement of try (24'a) also fails to imply (24b), while the infinitive complement of come (25'a) does imply (25b), and the negative of both fake-and and infinitive complements (25'cd) implies the negative of (25b).

Notice that it is important to distinguish the infinitive complements of (24',25') from the same surface strings where the infinitive is an underlying purpose clause. Purpose clauses are invariably non-factive.

4.3 Deleted Complements

The real-and readings of try and V, be sure and V require deleted complements on try, be sure, leading to the discourse coherence argument of (20). The real-and reading of the other V₁ requires at most a deleted directional adverb, which is to some extent supplied by the deictic quality of come and go. This makes it difficult to distinguish the real and fake-and readings, and

- (30) a. Go to visit Harry. ==> (unknown rule(s))
 b. Go and visit Harry. ==> (Fake-and Deletion)
 c. Go visit Harry.

Such an analysis has three advantages:

A. Sections 3 and 4 show that and visit Harry in (30b) has the semantics of a subordinate clause, a close paraphrase of (30a); thus the semantics will be roughly correct.

B. The presence of the infinitive will handle the V₂ part of the bare-stem condition.

C. And functions as an unmarked conjunction with a wide range of uses linking main and subordinate structures: and then, ">" as in "One more step and I shoot.", nice and warm, etc. It is therefore semantically plausible for it to function as a complementizer here.

Despite these advantages, any simple "to ==> and" analysis has both syntactic and semantic problems. First, negatives and auxiliary elements show different syntactic patterns:

- (31) a. *Go not to visit Harry. Try not to get lost.
 b. *Go not and visit Harry. *Try not and get lost.
 c. *Go to not visit Harry. ?OK Try to not visit Harry.
 d. ?OK Go and not visit H. ?OK Try and not visit H.
 (32) a. Try to have learned this by Monday.
 b. ?Try and have learned this by Monday.

These syntactic problems could perhaps be handled mechanically, but Donca Steriade points out two more serious semantic problems:

- (33) a. Be tall (see if I care). b. Go and be tall.
 c. Go be tall. d. *Go to be tall.
 (34) a. John didn't try to learn French, but living in Paris
 he just picked it up.
 b. ?John didn't try and learn French, but living in Paris
 he just picked it up.

(33abc) have the usual oddity of imperatives formed on non-self-controllables, but (*33d) is much worse. (34) shows that, while try to and try and are both non-factive, try and is much closer to being a necessary condition verb than try to. These semantic contrasts, one with try and better and one with try to better, suggest that the attempt to derive double-verb structures from an underlying infinitive complementizer is a mistake, and that we would do better to set up a structure with an ad hoc complementizer with different semantic properties.

7. Conclusion

This paper has two main points: First, we have showed that double-verb constructions are subject to a bare-stem condition, and that the statement of that condition tells us something about markedness in the English verb system. Second, we have showed that the surface and of the V and V structure is a fake and that cannot be derived from an underlying co-ordinate conjunction. In addition, we have speculated about the underlying structure of these double-

verb constructions, and suggested that they involve a complement structure that is not a simple surface replacement of an infinitive complementizer.

Footnotes

*We are grateful to Warren Cowgill and Rulon Wells, and especially to Donca Steriade; but none of them necessarily agrees with us.

1. Since completing the main work on this paper, we have found papers by Cohen (1968), Zwicky (1969), Faraci (1970), Levi (1971), and Shopen (1971). Our analysis differs significantly from any of theirs, but we have not attempted a systematic comparison here.

2. Some subjects report these are good, but a repetition test reveals a much-reduced and. See the discussion in section 4.

3. Some subjects report (*8b, *9b), while bad, to be detectably better than (*5b, *6b). We have no explanation for this partial phonological amnesty.

4. (13cd) have a marginal real-and reading, see section 4.1.

5. Donca Steriade ^{points out} that there is a principled distinction between "affix normally zero" (e.g. the present forms that are zero except with be) and "accidentally zero form of an affix normally non-zero" (e.g. the past participle of come). If such a distinction is needed independently, our argument for an unmarked present is greatly weakened.

Zwicky (1969) considers this same contrast, and proposes a rule of "Affix Deletion" for the present tense. We would reject this as ad hoc.

6. Some of these judgements may seem a bit delicate, but interviews by GC and Chris Clifford show that the data are actually very clean:

	OK	?OK	?*	*
John has <u>come</u> live with us.	3	1	1	22
Did Bill go visit Wells in the hospital?	26	1	—	—
If they <u>come</u> visit us in New Haven, ...	26	—	—	1

7. This conclusion may be relevant to the status of English as the notorious exception to the implicational universal that, if some agreement position is unmarked, it's the 3rd person singular.

8. We imply a generative model here, but the same argument could be reconstructed in an interpretive model.

9. Note that the status of (20bc), on the reading where you force a backwards anaphora with and..., is the same as cases like (i) where you force a backwards anaphora out of an overt subordinate clause:

- (i) John tried, although Bill had told him not to move
the piano alone.

10. Most of our disagreements with the earlier papers, we believe, result from their failure to make these distinctions.

11. (24d) is bad independently of the bare-stem violation, since negatives are blocked with be sure and V:

(ii) ?*John won't be sure and visit Harry. (only as an echo NEG)

12. At first glance it looks like a necessary-condition verb rather than a non-factive; but we believe this is an implicature rather than an entailment. Note the cancellation in (?34b).

13. Compare Shopen's discussion (1971:260). We reject his deixis analysis because the relevant sentences are equally bad with come, and blame the contrast between (*29b) and (29c) on the unexpected-event reading rather than on a difference in tightness of nexus.

14. We might point out that a rule mechanically inserting go, come, etc before bare stems fails on two counts: (A) The semantics will be a disaster. (B) Even if we take a strong autonomous-syntax position, one "purely syntactic" argument remains: pseudo-imperatives are presumably bare stems, but cannot be V₂:

(iii) *Go (and) fuck you!

References

- Cohen, David (1968) "Go fly a kite! and other suggestions," unpublished paper.
- Faraci, Robert (1970) "And as a verb-phrase complementizer," 1st NELS
- Levi, Judith N. (1971) "Nonconjunctive and's," unpublished paper
- Shopen, Tim (1971) "Caught in the act," 7th CLS: 254-63
- Zwicky, Arnold (1969) "Phonological Constraints in Syntactic Descriptions," Papers in Linguistics 1:411-63